Kevin Wang Ph.D.

Data Scientist and Statistician

US: (+1) 619-215-3725
AU: (+61) 404-955-255
kevinwangstats@gmail.com
Portfolio: kevinwangstats.com

Experience

Now **Senior Data Scientist**, *Illumina*

2021 Nov
- Serving as the bioinformatics lead for the TruSight Oncology 500 ctDNA product, driving comprehensive scientific decisions while partnering directly with external strategic accounts to guide product roadmap and planning.
- Facilitating cross-functional collaboration among R&D scientists, software developers, regulatory authorities, and pharmaceutical partners to secure development and compliance milestones across both RUO portfolios and highly regulated IVD products in the US and EU.
- Advanccing data infrastructure and clinical data science practices supporting both internal R&D teams and external customers by upgrading sequencing storage with curated database architecture, as well as deploying an AI-powered ctDNA knowledge base and tools in Python and R, reducing customer troubleshooting turnaround from days to minutes and data verification from months to hours.
- Leading bioinformatics development for multiple pioneering cancer companion diagnostics programs, overseeing high-value data deliverables which directly contribute to multimillion-dollar contracts and regulatory milestones.
- Develop machine learning algorithms to analyze large-scale genomics datasets across oncology and prenatal products, uncovering clinically-actionable insights and authoring a high-impact publication highlighting clinical benefits in blood cancer.

2021 Oct **Statistician**, *CSL Behring*

2020 Jul
- Direct contributions to the R&D and manufacturing of high-value pharmaceutical products like the COVID-19 vaccines and treatments during the pandemic.
- Designed and formalized statistical process control techniques to standardize practices in the manufacturing and quality control laboratory. These tools were deployed through interactive dashboards which had a wide organizational impact.
- Provided actionable insights to multiple pharmaceuticla products which mitigated significant regulatory risks and achieved significant financial savings, with some up to multimillion dollars production batches.

2020 Jun **Research Associate**, *University of Sydney*

2019 Sep
- Developed an interpretable machine learning framework for predicting patient clinical outcomes using omics data. Melanoma was a key focus of this research and the key biomarkers were validated and implemented on a cost-effective assay.
- `R` software packages associated with the research were developed and maintained in a reproducible environment.
- Visiting scholar at Cornell University for 2 months. Orchestrated operations, designed contents and hosted a workshop in open-source single-cell analytics leveraging cloud computing.
- Published peer-reviewed articles in top-tier journals, including PNAS, Nature Digital Medicine and Nature Computational Science.

2019 Mar **Postgraduate Teaching Fellow**, *University of Sydney*

2016 Mar
- Delivered lectures (200+ students) and tutorials, covering 15 different courses, including statistics, mathematics and data science. Achieved 85% satisfactory rating for end-of-course surveys.
- Authored and designed course content at both the undergraduate and postgraduate levels.
- Mentored and trained junior statistics tutors.

Education

2016–2020 **Doctor of Philosophy in Statistical Bioinformatics**, *University of Sydney*

- Developed machine learning methods to solve statistical challenges in patient clinical outcome predictions and thereby enabling personalised medicine.
- Real-world biomedical data from collaborators were used to motivate and articulate important challenges in current statistical research. The value of this thesis is the synthesis of statistics, clinical implementation and practical deployment.

2012-2015 **Bachelor of Science (Adv. Mathematics) (Hon. I)**, *University of Sydney*

- Major in statistics and financial mathematics.
- The Honours thesis examined functional MRI data and the inferred brain connectivity networks through the use of different statistical association measures.

Skills

AI & Machine learning — Extensive experience designing and deploying AI solutions, machine learning algorithms, and predictive models across research and regulated production environments. Expertise in predictive modeling for high-dimensional genomics data, AI-powered knowledge bases for rapid technical troubleshooting, and extracting clinically-actionable insights from large-scale biomedical datasets.

`python` — Developed computational modules for regulated products at Illumina to support clinical studies. Highly proficient in writing scripts and notebooks to solve analytical problems for stakeholders. Experience in using and applying `pandas`, `numpy`, `scikit-learn`, `keras`, `pytorch`, and `openai` for a range of projects.

`R` — Highly proficient in scripting and developing packages. Strong ability to use `tidyverse` and `shiny` for the data ETL process to produce visualisations and presentable insights. Author and maintainer of several open-source `R` packages, including `APES`, `learningtower`, `mcvis` and `scMerge`.

`git` & CI/CD & docker — Highly proficient in a reproducible environment that supports business operations and software development requirements. Experience in holding reproducible workshops with excellent participant feedback.

Applied statistics — Highly experienced at the research level and consulting level. Ability to transform business and cross-disciplinary problems into a statistical framework and deliver solutions. Strong experience with model selection to extract explainable insights from data.

Other skills — SQL, cloud and dashboard development for business use.

Volunteering

2021-Present **Mentor**, *Monash University*

Supervised and mentored three Master of Data Science students with the students successfully completed their research projects. Developed high-quality R-CRAN package and published academic journal article.

2021-2023 **Councillor**, *Statistical Society of Australia*

Served as the Assistant Secretary and Communication Officer in the NSW and Victorian Branch of the Society. Organised public events, designed automated management systems, responsible for member communications and social media account maintenance.

Publications

1 **Wang, K. Y. X.**, Pupo, G.M., Tembe, V., Patrick, E., Strbenac, D., Schramm, S.J., Thompson, J.F., Scolyer, R.A., Muller, S., Tarr, G. and Mann, G.J., Yang, J.Y.H. **2022**. Cross-Platform Omics Prediction procedure: a statistical machine learning framework for wider implementation of precision medicine. Nature Digital Medicine, 5(1) 1-10.

2 Kim J.H., **Wang, K. Y. X.**, Chen, C., Lin, Y., Tam, P.P.L., Lin, D.M., Yang, J.Y.H., & Yang, P. **2021**. Cepo uncovers cell identity through differential stability. *Nature Computational Science*, **1** 784–790.

3 Schafer, S., **Wang, K. Y. X.**, Sundling, F., Yang, J.Y.H., & Liu, A. **2021**. Modelling maternal and perinatal risk factors to predict poorly controlled childhood asthma. *PLOS ONE* 16(5): e0252215.

4 Lin, C., **Wang, K. Y. X.**, & Mueller, S. **2020**. mcvis: A new framework for collinearity discovery, diagnostic and visualization. *Journal of Computational and Graphical Statistics*, 1-13.

5 Hewavisenti, R., Ferguson, A., **Wang, K. Y. X.**, Jones, D., Gebhardt, T., Edwards, J., Zhang, M., Britton, W., Yang, J., Hong, A., & Palendira, U. **2020**. CD103+ tumour-resident CD8+ T cell numbers underlie improved patient survival in oropharyngeal squamous cell carcinoma. *Journal for ImmunoTherapy of Cancer*, 8:e000452.

6 **Wang, K.Y.X.**, Tarr, G., Yang, J.Y.H., Mueller, S. **2019**. Fast and approximate exhaustive variable selection for generalised linear models with APES, Invited paper to *Australia & New Zealand Journal of Statistics*, 61 (4) 445-465.

7 Lin, Y., Ghazanfar, S., **Wang, K.Y.X.**, Gagnon-bartsch, J.A., Lo, K.K., Han, Z., Ormerod, J.T., Speed, T.P., Yang, P., Yang, J.Y.H. **2019**. scMerge: Leveraging factor analysis, stable expression and pseudo-replication to merge multiple single-cell RNA-seq data, *Proceedings of the National Academy of Sciences of the United States of America*, 116 (20) 9775-9784.

8 Pires da Silva, I., **Wang, K.Y.X.**, Wilmott, J.S., Holst, J., Carlino, M.S., Park, J.J., Quek, C., Wongchenko, M., Yan, Y., Mann, G., Johnson, D.B., McQuade, J.L., Rai, R., Kefford, R.F., Rizos, H., Scolyer, R.A., Yang, J.Y.H., Long, G. V, Menzies, A.M. **2019**. Distinct molecular profiles and immunotherapy treatment outcomes of V600E and V600K BRAF-mutant melanoma. *Clinical Cancer Research*, 25 (4) 1272-1279.

9 **Wang, K.Y.X**., Menzies, A.M., Silva, I.P., Wilmott, J.S., Yan, Y., Wongchenko, M., Kefford, R.F., Scolyer, R.A., Long, G. V, Tarr, G., Mueller, S., Yang, J.Y.H. **2019**. bcGST - an interactive bias-correction method to identify over-represented gene-sets in boutique arrays. *Bioinformatics*, 35 (8) 1350-1357.

10 Strbenac, D., **Wang, K.Y.X.**, Wang, X., Dong, J., Mann, G.J., Mueller, S., Yang, J.Y.H. **2019**. Melanoma Explorer: a web application to allow easy reanalysis of publicly available and clinically-annotated melanoma omics datasets. *Melanoma Research*.